

AMENDMENT NO. 1 JANUARY 2024
TO
IS 15844 (PART 1) : 2023 SPORTS FOOTWEAR
PART 1 GENERAL PURPOSE
(First Revision)

(Page 1, clause 3.3) — Insert the following new clause after 3.3:

‘3.4 Insock

Removable or non-removable footwear component used to cover part or all of the insole.

NOTE — “Non-removable” means that the insock cannot be removed without being damaged.’

(Page 1, clause 4.2) — Substitute ‘Category and Type’ for ‘Construction’.

(Page 1, clause 4.2.1) — Substitute the following for existing:

‘4.2.1 The sports footwear: General purpose shall be of the following two categories on the basis of their usage:

- a) For kids (standard English sizes for children as per IS 1638); and
- b) For adults (for all other sizes as per IS 1638).

Further, sports footwear: General purpose for adults shall be of two types namely, Type 1 and Type 2.’

(Page 1, clause 4.3) — Substitute the following for existing clause:

‘5 REQUIREMENTS OF WHOLE FOOTWEAR

The sports footwear: General purpose shall conform to the requirements as specified in Table 1.

Table 1 Requirements of Whole Footwear

(Clause 5)

SI No.	Characteristic	Requirement			Methods of Test, Ref to
		Kids	Adults		
			Type 1	Type 2	
(1)	(2)	(3)	(4)	(5)	(6)
i)	Insole, if used	Cannot be removed without damaging the footwear	Cannot be removed without damaging the footwear	Cannot be removed without damaging the footwear	–
ii)	Toe spring height, mm, <i>Min</i>	2	2	2	Fig. 1 of Annex B
iii)	Bond performance, N/mm, <i>Min</i> a) Upper to sole/mid sole/ composite sole	1.0 1.0 (in case of material tear)	2.5 2.0 (in case of material tear)	1.5 1.0 (in case of material tear)	Annex C

Price Group 6

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SI No.	Characteristic	Requirement			Methods of Test, Ref to
		Kids	Adults		
			Type 1	Type 2	
(1)	(2)	(3)	(4)	(5)	(6)
	b) Interlayer bond strength (applicable only for multilayer pasted stuck-on bottom sole)	1.0 1.0 (in case of material tear)	2.5 2.0 (in case of material tear)	1.5 0.5 (in case of material tear)	
iv)	Energy absorption at seat region, J, <i>Min</i>	–	15	–	IS 15298 (Part 1)
v)	Complete shoe flexing,	–	At an angle of 45 degrees at 100 000 cycles, no change in upper (for example, crack, thread breakage, fabric broken), sole cracking, bond opening shall occur	At an angle of 45 degrees at 60 000 cycles, no change in upper (for example, crack, thread breakage, fabric broken), sole cracking, bond opening shall occur	IS 8085 (Part 16)/ ISO 24266
vi)	Washability (optional requirement)	–	There shall not be any bond opening, coating separation, and grey scale rate shall be minimum 3 grey scale	–	IS 8085 (Part 15)/ ISO 19954
vii)	Slip resistance, coefficient of friction (COF), <i>Min</i> At both dry and wet condition on clay tiles (not applicable for leather sole) (to be carried out on quarry tiles)	–	0.3	–	Annex E
viii)	Abrasion resistance ¹ (volume loss), mm ³ , <i>Max</i> a) Cellular sole (load 5 N) b) Solid sole (load 10 N)	700 400	500 350	700 400	IS 3400 (Part 3)
(x)	Chemical requirements ²	All the components of sport footwear	All the components of sport footwear	All the components of sport footwear	Table 1 of IS 17011

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SI No.	Characteristic	Requirement			Methods of Test, Ref to
		Kids	Adults		
			Type 1	Type 2	
(1)	(2)	(3)	(4)	(5)	(6)
		which are in direct contact of foot shall comply with Table 1 of IS 17011 for critical substances Category I and Category II as specified under 3.6 of IS 17011	which are in direct contact of foot shall comply with Table 1 of IS 17011 for critical substances Category I and Category II as specified under 3.6 of IS 17011	which are in direct contact of foot shall comply with Table 1 of IS 17011 for critical substances Category I and Category II as specified under 3.6 of IS 17011	
(xi)	Hydrolysis resistance, <i>Max</i> (applicable for polyester PU sole only)	After 25 000 flexes 6.0 mm cut growth	After 150 000 flexes 6.0 mm cut growth	After 50 000 flexes 6.0 mm cut growth	IS 15298 (Part 1)
(xii)	Lead content (as Pb) ³ ppm, <i>Max</i> (applicable only for PVC)	2	2	–	IS 12240 (Part 5)
NOTES ¹ Solid, density more than 0.90 g/cm ³ ; and Cellular, density less than 0.90g/cm ³ . ² In case of Type 2 and kids, the chemical requirements shall be tested after 31 st December 2024. ³ To be tested only If PVC is used in manufacturing of sports shoes.					

6 OPTIONAL REQUIRMENTS FOR COMPONENTS

The tests for the various components used in the manufacture of sports footwear: General purpose as given in **4.4, 4.5, 4.6, 4.7, 4.8, 4.9, 4.10, 4.11, 4.12** and **4.13** shall be optional and may be tested. The characteristics to be tested and the requirements specified against them are given for guidance purpose only.’

(Page 2, Table 1) — Substitute the following table for the existing table:

Table 2 Upper Material – Leather

(Clause 4.4)

SI No.	Characteristic	Requirements			Method of Test, Ref to
		Kids	For Adults		
			Type 1	Type 2	
(1)	(2)	(3)	(4)	(5)	(6)
i)	Tear strength (average force), <i>Min</i> , N	—	50	50	IS 5914 (Part 5/Sec 2)/ ISO 3377-2

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SI No.	Characteristic	Requirements			Method of Test, Ref to
		Kids	For Adults		
			Type 1	Type 2	
(1)	(2)	(3)	(4)	(5)	(6)
ii)	Colour fastness rubbing (to and fro) grey scale rate, <i>Min</i> a) Dry: (Rubbing) Rating: (Marring and staining) b) Wet: (Rubbing) Rating: (Marring and staining)	— —	150 3 grey scale 50 3 grey scale	— —	IS 6191 (Part 4)
iii)	a) Water vapour permeability, mg/cm ² h, <i>Min</i> b) Water vapour coefficient, mg/cm ² , <i>Min</i>	— —	0.8 15	— —	IS 15298 (Part 1) IS 15298 (Part 1)
iv)	Stitch tear strength (double hole), N/mm, <i>Min</i>	—	50	—	LP 8 of IS 5914
v)	Colour fastness to light (optional) grey scale rate, <i>Min</i> Rating: (Marring)	—	4	—	IS/ISO 105B02

(Page 3, Table 2) — Substitute the following table for the existing table:

Table 3 Upper Material –Material Other Than leather

(Clause 4.4)

SI No.	Characteristic	Requirements			Method of Test, Ref to
		For Kids	For Adults		
			Type 1	Type 2	
(1)	(2)	(3)	(4)	(5)	(6)
i)	Breaking strength, N/mm, <i>Min</i>	—	6.0	—	IS 7016 (Part2) Method 1

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SI No.	Characteristic	Requirements			Method of Test, Ref to
		For Kids	For Adults		
			Type 1	Type 2	
(1)	(2)	(3)	(4)	(5)	(6)
ii)	Elongation at break, percent <i>Min</i> a) Tighter direction b) Stretch direction	– –	7 15	– –	
iii)	Tear strength, N, <i>Min</i> a) For fabric b) For all other materials	15 20	20 30	20 30	IS 15298 (Part 1)
iv)	Flexing resistance, (applicable for coated fabrics) a) Dry condition b) Wet condition	– –	No damage to coating/ lamination till 100 000 flexes No damage to coating/ lamination till 20 000 flexes	– –	IS 8085 (Part 20)/ ISO 17694
v)	a) Water vapour permeability, mg/cm ² .h, Min b) Water vapour coefficient, mg/cm ² Min (applicable for textile upper)	– –	0.8 15.0	– –	IS 15298 (Part 1)
vi)	Hydrolysis resistance (after ageing at 70 °C and 95 percent RH for 7 days) (applicable only for material having PU)	No damage/ crack to coating/ lamination till 50 000 flexes	No damage/crack to coating/ lamination till 100 000 flexes	No damage/crack to coating/ lamination till 75 000 flexes	IS 15298 (Part 1)

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SI No.	Characteristic	Requirements			Method of Test, Ref to
		For Kids	For Adults		
			Type 1	Type 2	
(1)	(2)	(3)	(4)	(5)	(6)
vii)	Bursting strength, kg/cm ² , <i>Min</i> (additional test required for knitted fabrics)	—	8	—	IS 1966 (Part 2)

(Page 4, Table 3) — Substitute the following table for the existing table:

Table 4 Lining Material – All Materials

(Clause 4.5)

SI No.	Characteristic	Requirement			Method of Test, Ref to
		For Kids	For Adults		
			Type 1	Type 2	
(1)	(2)	(3)	(4)	(5)	(5)
i)	Tear strength, N, <i>Min</i>	—	15	—	IS 15298 (Part 1)
ii)	Abrasion resistance, cycles				IS 15298 (Part 1)
	a) Dry condition	Shall not develop any hole before 6 400 cycles	Shall not develop any hole before 12 800 cycles	Shall not develop any hole before 6 400 cycles	
	b) Wet condition	Shall not develop any hole before 3 200 cycles	Shall not develop any hole before 6 400 cycles	Shall not develop any hole before 3 200 cycles	
iii)	Colour fastness to rubbing (crocking), grey scale rate with 10 rubs on dry or wet, <i>Min</i> Rating: (Staining)	—	3	—	IS/ISO 105-X12
iv)	Colour fastness to perspiration	—	3	—	IS 6191/ISO 11641

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SI No.	Characteristic	Requirement			Method of Test, Ref to
		For Kids	For Adults		
			Type 1	Type 2	
(1)	(2)	(3)	(4)	(5)	(5)
	(contact method) grey scale rate, <i>Min</i> Rating: (Staining)				
v)	a) Water vapour permeability, mg/cm ² h, <i>Min</i> b) Water vapour coefficient, mg/cm ² , <i>Min</i> (excluding coated fabric)	— —	2.0 20.0	— —	IS 15298 (Part 1) IS 15298 (Part 1)

(Page 5, Table 4) — Substitute the following table for the existing table:

Table 5 Insole — All Materials

(Clause 4.6)

SI No.	Characteristic	Requirement			Method of Test, Ref to
		For Kids	For Adults		
			Type 1	Type 2	
(1)	(2)	(3)	(4)	(5)	(6)
i)	Thickness, mm, <i>Min</i>	–	0.8	–	IS 15298 (Part 1)
ii)	Flexing index, <i>Min</i> (only applicable for cellulose board)	–	2.7	–	Annex ‘F’
iii)	Abrasion resistance, at 400 cycles	–	No surface tearing	–	IS 15298 (Part 1)
iv)	a) Water absorption, mg/cm ² , <i>Min</i>	–	60	–	IS 15298 (Part 1)
	b) Water desorption, percent, <i>Min</i>	–	70	–	IS 15298 (Part 1)

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(Page 5, Table 5) — Substitute the following table for the existing table:

Table 6 In-sock- All Materials

(Clause 4.7)

Sl No.	Characteristic	Requirement			Method of Test, Ref to
		For Kids	For Adults		
			Type 1	Type 2	
(1)	(2)	(3)	(4)	(5)	(6)
i)	Thickness, mm, <i>Min</i> (including foam)	–	1.5	–	IS 15298 (Part 1)
ii)	Abrasion resistance, cycles				
	a) Dry condition	Shall not develop any hole before 6 400 cycles	Shall not develop any hole before 12 800 cycles	Shall not develop any hole before 6 400 cycles	IS 15298 (Part 1)
	b) Wet condition	Shall not develop any hole before 3 200 cycles	Shall not develop any hole before 6 400 cycles	Shall not develop any hole before 3 200 cycles	IS 15298 (Part 1)
iii)	a) Water absorption, mg/cm ² , <i>Min</i>	–	10	–	IS15298 (Part 1)
	b) Water desorption, percent, <i>Min</i>	–	50	–	IS 15298 (Part 1)

(Page 6, Table 6) — Substitute the following table for the existing table:

Table 7 Out Sole (All Material)

(Clause 4.8)

SI No.	Characteristic	Requirement			Method of Test, Ref to, IS
		For Kids	For Adults		
			Type 1	Type 2	
(1)	(2)	(3)	(4)	(5)	(6)
i)	Flexing resistance Belt method at 90 diameter Mandrel, or Bennewart method, cut growth, <i>Max</i>	— —	No crack at 30 000 flexes 6 mm	— —	IS 8085 (Part 4) Annex D

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SI No.	Characteristic	Requirement			Method of Test, Ref to, IS
		For Kids	For Adults		
			Type 1	Type 2	
(1)	(2)	(3)	(4)	(5)	(6)
ii)	Hydrolysis resistance, cut growth after flexes, in mm, <i>Max</i> (Only for PU out sole)	After 25 000 flexes 6.0 mm cut growth	After 150 000 flexes 6.0 mm cut growth	After 50 000 flexes 6.0 mm cut growth	IS 15298 (Part 1)

(Page 6, Table 7) — Substitute the following table for the existing table:

Table 8 Mid Sole

(Clause 4.9)

SI No.	Characteristic	Requirement			Method of Test, Ref to, Annex
		For Kids	For Adults		
			Type 1	Type 2	
(1)	(2)	(3)	(4)	(5)	(6)
i)	Compression set, percent, <i>Max</i> a) For EVA b) For all other materials	 65 50	 60 50	 65 50	Annex G
ii)	Split tear strength, kg/25 mm, <i>Min</i>	—	1.5	—	Annex H
iii)	Heat shrinkage, percent, <i>Max</i>	—	5.0	—	Annex J

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(Page 7, Table 8) — Substitute the following table for the existing table:

Table 9 Counter Stiffener

(Clause 4.10)

Sl. No.	Characteristic	Requirement			Method of Test, Ref to Annex of IS 7554
		For Kids	For Adults		
			Type 1	Type 2	
(1)	(2)	(3)	(4)	(5)	(6)
i)	First dry collapsing load, hardness, N, <i>Min</i>	–	15	–	A-3.2.6
ii)	Resilience, percent, <i>Min</i>	–	50	–	A-3.4
iii)	Moisture resistance, percent, <i>Min</i>	–	40	–	A-3.5
iv)	a) Initial dry area shape retention, percent, <i>Min</i>	–	55	–	A-3.1
	b) Area shape retention after 10th collapse, dry, percent, <i>Min</i>	–	45	–	A-3.3

(Page 7, Table 9) — Substitute the following table for the existing table:

Table 10 Velcro Fastener

(Clause 4.11)

SI No.	Characteristic	Requirement			Method of Test, Ref to
		For Kids	For Adults		
			Type 1	Type 2	
(1)	(2)	(3)	(4)	(5)	(6)
i)	Peel strength, N/mm, <i>Min</i>				IS 8085 (Part 18)/ ISO 22777
	a) Initial	—	0.10	—	
	b) After 5 000 wear cycles	—	0.08	—	
ii)	Shear strength, kPa, <i>Min</i>				ISO 22776
	a) Initial	—	75	—	
	b) After 5 000 wear cycles	—	65	—	

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(Page 8, Table 10) — Substitute the following table for the existing table:

Table 11 Elastic Tape

(Clause 4.12)

Sl No.	Characteristic	Requirement			Method of Test, Ref to
		For Kids	For Adults		
			Type 1	Type 2	
(1)	(2)	(3)	(4)	(5)	(6)
i)	Limit of useful extension, percent, <i>Min</i>	—	90	—	IS 8085 (Part 12)/ ISO 10765

(Page 8, Table 11) — Substitute the following table for the existing table:

Table 12 Fasteners (Lace/Buckle/Eyelet/D-Ring/Ski-Hook/Metal Trims)

(Clause 4.13)

SI No.	Characteristic	Requirement			Method of Test, Ref to
		For Kids	For Adults		
			Type 1	Type 2	
(1)	(2)	(3)	(4)	(5)	(6)
i)	Breaking strength of lace, N, <i>Min</i>	–	250	–	Annex K
ii)	Lace tag strength, N, <i>Min</i>	–	100	–	Annex L
iii)	Lace to lace abrasion,	–	No breakage or damage before 4 000 cycles	–	IS 8085 (Part 17)/ ISO 22774
iv)	Colour fastness to water (for lace) (contact method) grey scale rating, <i>Min</i>	–	3	–	IS 6191 (Part 2) or IS/ISO 105-E01
v)	Strength of buckle/ski hook, N, <i>Min</i>	–	250	–	Annex M
vi)	Corrosion resistance (metal parts)	–	Not worse than slight uniform change	–	IS 17098 (both Method 1 and Method 2)

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(Page 9, clause **4.14**) — Delete.

(Page 9, clause **5**) — Substitute the following for the existing and renumber the subsequent clauses:

‘7 MARKING AND PACKING

7.1 The sports footwear shall be marked legibly with the following:

- a) Size;
- b) Type; and
- c) Identification of the source of manufacturer or brand name.

NOTE — Kids sports shoes shall be marked as ‘0’.

(Page 9, clause **5.3**, first line) — Substitute ‘may’ for ‘shall’.

(Page 11, Annex B, Title) — Substitute the following for existing:

‘ANNEX B

[clause 5, Table 1, Sl No.(ii)].

TOE SPRING HEIGHT’

(Page 11, Annex C, Title) — Substitute the following for existing:

‘ANNEX C

[clause 5, Table 1, Sl No. (iii)]

ADHESION TEST’

(Page 11, **C-3.7**) — Delete.

(Page 21, clause **E-6**, second line) — Substitute ‘27 °C ± 2 °C and (65 ± 5) percent relative humidity’ for ‘23 °C ± 2 °C and 50 percent ± 5 percent RH’.

(Page 21, clause **E-6**, fifth line) — Substitute ‘27 °C ± 2 °C’ for ‘23 °C ± 2 °C’.

(Page 25, clause **G-1.4.5**, third line) — Substitute ‘27 °C ± 2 °C’ for ‘(20 ± 2) °C’.

(Page 30, clause **K-3.4**, second line) — Substitute ‘27 °C ± 2 °C’ for ‘23 °C ± 2 °C’.

(Page 31, clause **K-4.3**, second line) — Substitute ‘27 °C ± 2 °C and (65 ± 5) percent relative humidity’ for ‘23 °C ± 2 °C/50 percent ± 4 percent RH’.

(Page 31, clause **K-5.11**, second line) — Substitute ‘27 °C ± 2 °C’ for ‘(23 ± 2) °C’.

(Page 41, clause **P-6**, third line) — Substitute ‘27 °C ± 2 °C’ for ‘(23 ± 2) °C’.

(Page 42, clause **P-9.1**, second para, fourth line) — Substitute ‘27 °C ± 2 °C and (65 ± 5) percent relative humidity’ for ‘23 °C ± 2 °C and 50 percent ± 5 percent RH’.

(Page 42, clause **P-10.1.1**, second line) — Substitute ‘27 °C ± 2 °C and (65 ± 5) percent relative humidity’ for ‘23 °C ± 2 °C and 50 percent ± 5 percent RH’.